EC2 vs Fargate Cost Analysis — EMR-Bear EKS Migration

Prepared: 2026-03-22 **Context:** DigitalOcean → AWS EKS migration, Multi-AZ + Multi-Region DR **Primary Region:** us-west-2 (co-located with existing AWS IoT, DynamoDB, Bedrock, S3) **DR Region:** us-east-1 (cold standby)

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1. Current Workload Profile

Component	Count	Current DO Spec	Notes
Tenant web pods (Puma/Rails)	~200	1-2 replicas per tenant	`bear-app`
DelayedJob worker pods	~167	1 replica per tenant	Background processing
Infrastructure pods	~50	Various	bear-services, x12, xero, trip-form-filler, stats, cron, registry
Prex healthcare pods	~13	Various	cerner, surescripts, idology, saaspass
Monitoring (DaemonSets)	79 (1/node)	node-exporter, etc.	**Cannot run on Fargate**
StatefulSets	4	Memcached, Redis, Prometheus	**Cannot use EBS on Fargate**
Total pods	**~1,141**		

Current DO Node Pools (what we're replacing)

Pool	Nodes	DO Spec	Total vCPU	Total RAM
`standard-pool`	40	s-6vcpu-16gb	240	640 GB
`delayed-job`	38	s-6vcpu-16gb	228	608 GB
`monitoring`	1	s-2vcpu-4gb	2	4 GB
Total	**79**		**470**	**1,252 GB**

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2. EC2 Cost Breakdown

2.1 Instance Mapping (from migration plan)

Pool	AWS Instance	Spec	Node Count
`standard-pool`	m6i.2xlarge	8 vCPU / 32 GB	40
`delayed-job`	m6i.xlarge	4 vCPU / 16 GB	38

`monitoring`	t3.medium	2 vCPU / 4 GB	1

> ***Note:*** *The migration plan oversizes the standard pool (32 GB vs DO's 16 GB). A right-sized alternative is discussed in Section 2.3.*

2.2 EC2 Pricing Tiers (monthly, us-west-2)

Pricing Model	standard-pool (40x m6i.2xlarge)	delayed-job (38x m6i.xlarge)	monitoring (1x t3.medium)	EKS Control	**Monthly Total**	**Annual Total**
On-Demand	$11,213	$5,326	$30	$73	**$16,642**	**$199,704**
1yr RI (No Upfront)	$6,920	$3,287	$19	$73	**$10,299**	**$123,588**
1yr Savings Plan	$7,849	$3,728	$21	$73	**$11,671**	**$140,052**
3yr RI (All Upfront)	$4,480	$2,128	$12	$73	**$6,693**	**$80,316**
Hybrid: RI standard + Spot delayed-job	$6,920	$1,634*	$19	$73	**$8,646**	**$103,752**

*Spot pricing for m6i.xlarge estimated at ~$0.058/hr (70% off on-demand). DelayedJob workers are interruption-tolerant — ideal Spot candidates.

2.3 Right-Sized EC2 Alternative

The standard pool uses m6i.2xlarge (32 GB) but DO only had 16 GB per node. If profiling shows 16 GB is sufficient:

Pool	Instance	Nodes	1yr RI/mo	3yr RI/mo
`standard-pool`	m6i.xlarge (4 vCPU / 16 GB)	60*	$5,190	$3,360
`delayed-job`	m6i.xlarge (4 vCPU / 16 GB)	38	$3,287	$2,128
`monitoring`	t3.medium	1	$19	$12
EKS Control Plane	—	—	$73	$73
Total		**99**	**$8,569**	**$5,573**
Annual			**$102,828**	**$66,876**

*60 nodes needed to match 240 vCPU (60 × 4 = 240) from DO's standard pool.

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3. Fargate Cost Breakdown

3.1 Fargate Pricing (us-west-2)

Resource	Rate
vCPU	$0.04048/hr
Memory (per GB)	$0.004445/hr

3.2 Pod-Level Resource Sizing

Pod Type	Count	vCPU	Memory	Cost/pod/hr	Monthly Total
Tenant web (Rails/Puma)	200	1	3 GB	$0.05381	$7,857
DelayedJob workers	167	0.5	1 GB	$0.02469	$3,009
bear-services (+ HDS sidecar)	1	2	4 GB	$0.09874	$72
bear-services-nginx	1	0.5	1 GB	$0.02469	$18
bear-x12	2	0.5	1 GB	$0.02469	$36
bear-xero	1	0.5	1 GB	$0.02469	$18
trip-form-filler	3	0.5	1 GB	$0.02469	$54
bear-stats + bear-github	2	0.5	1 GB	$0.02469	$36
Prex services	13	0.5	1 GB	$0.02469	$234
Fargate subtotal	**390**				**$11,334**

3.3 Components That CANNOT Run on Fargate

Component	Why	Solution	Est. Monthly Cost
DaemonSets (10 total: node-exporter, etc.)	Fargate doesn't support DaemonSets	EC2 node group + sidecar pattern	$150 (3x t3.medium)
Memcached StatefulSet	Fargate doesn't support EBS volumes	ElastiCache Memcached (2 nodes)	$120
Redis StatefulSet	Same — no EBS	ElastiCache Redis (1 node)	$60
Prometheus StatefulSet (50Gi PVC)	Same — no EBS	EC2 node for monitoring OR managed Prometheus	$200
Docker Registry (5Gi mem request)	Can run on Fargate but better to replace	**Use ECR instead** ($0)	$0
Non-Fargate subtotal			**$530**

3.4 Fargate Total Cost by Pricing Tier

Pricing Model	Fargate Pods	Non-Fargate Infra	EKS Control	**Monthly Total**	**Annual Total**
On-Demand	$11,334	$530	$73	**$11,937**	**$143,244**
1yr Savings Plan (20% off)	$9,067	$530	$73	**$9,670**	**$116,040**
3yr Savings Plan (45% off)	$6,234	$530	$73	**$6,837**	**$82,044**

3.5 Fargate Limitations for This Workload

Limitation	Impact	Severity
No DaemonSets	10 DaemonSets (monitoring agents) need EC2 or sidecar refactoring	Medium
No EBS volumes	Prometheus, Memcached, Redis StatefulSets can't run on Fargate	Medium
No privileged containers	Some monitoring/debugging tools won't work	Low
1 ENI per pod	1,141 pods = 1,141 ENIs; VPC subnet must be /16 or larger	High — plan subnets carefully
Max 4 vCPU / 30 GB per pod	OK for this workload (no pod needs more)	None
Cold starts	New pods take 30-60s longer than EC2 (image pull + microVM init)	Low
No node-level SSH	Cannot SSH into nodes for debugging	Low (use `kubectl exec`)

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4. Head-to-Head Comparison

4.1 Cost Summary Table

Scenario	Monthly	Annual	vs. Cheapest
EC2 On-Demand	$16,642	$199,704	+149%
Fargate On-Demand	$11,937	$143,244	+79%
EC2 1yr RI (No Upfront)	$10,299	$123,588	+54%
EC2 1yr Savings Plan	$11,671	$140,052	+75%
Fargate 1yr Savings Plan	**$9,670**	**$116,040**	+45%

EC2 Hybrid (RI + Spot workers)	$8,646	$103,752	+30%
EC2 Right-sized 1yr RI	$8,569	$102,828	+28%
Fargate 3yr Savings Plan	$6,837	$82,044	+2%
EC2 3yr RI (All Upfront)	$6,693	$80,316	+0.2%
EC2 Right-sized 3yr RI	**$5,573**	**$66,876**	**Cheapest**

4.2 Total Cost of Ownership (including operational overhead)

Factor	EC2	Fargate
OS patching / AMI updates	You manage (monthly)	AWS manages
Node scaling / right-sizing	You manage (Karpenter/CA)	Automatic per-pod
Security hardening (CIS benchmarks)	You manage	AWS manages
Kubernetes version upgrades (nodes)	You coordinate	AWS handles Fargate platform
Monitoring node health	You manage	Not applicable
Estimated DevOps time	0.25-0.5 FTE (~$30-60K/yr)	Near zero
True annual cost (1yr commit)	**$123,588 + $45K ops = ~$169K**	**$116,040 + $5K ops = ~$121K**
True annual cost (3yr commit)	**$80,316 + $45K ops = ~$125K**	**$82,044 + $5K ops = ~$87K**

> ***Key insight:*** *Fargate appears ~30% more expensive on raw compute, but when you factor in operational overhead, the gap closes significantly. For a team without dedicated K8s node management expertise, Fargate can actually be cheaper in total cost of ownership.*

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5. Multi-AZ Architecture Cost

5.1 Option A: Preferred Zone + Warm Standby (Andres's preferred, from screenshot)

```
us-west-2a (PRIMARY)                    us-west-2b (STANDBY)
┌──────────────────────────┐            ┌──────────────────────┐
│  Full EKS Cluster        │            │  2-3 Warm Spare      │
│  (all ~1,141 pods)       │            │  Nodes (idle)        │
│                          │            │                      │
│  RDS Primary             │──Multi-AZ──▶ RDS Standby          │
│  (all instances)         │  (auto)    │  (auto-failover)     │
│                          │            │                      │
│  ElastiCache Primary     │            │  ElastiCache Replica  │
└──────────────────────────┘            └──────────────────────┘
```

Cost Component	Monthly	Annual

Primary AZ compute (see Section 2 or 3)	(base cost)	(base cost)
Warm spare nodes (2-3 × m6i.xlarge RI)	$173 - $260	$2,076 - $3,120
RDS Multi-AZ standby (doubles RDS compute)	(included in RDS pricing below)	
Cross-AZ data transfer	~$42	~$500
Additional cost over single-AZ	**~$250/mo**	**~$3,000/yr**

AZ Failure RTO: 2-5 minutes (RDS auto-failover: 60-120s, pods reschedule to standby + autoscaler scales up: 5-15 min for full capacity) **Data Loss:** 0

5.2 Option B: Active-Active 50/50 Multi-AZ (Ramesh's recommendation)

```
us-west-2a (ACTIVE)                    us-west-2b (ACTIVE)
┌──────────────────────────┐           ┌──────────────────────────┐
│  50% EKS Pods            │◄──Cross-AZ──► 50% EKS Pods          │
│  (~570 pods)             │   traffic   │  (~570 pods)           │
│                          │             │                        │
│  RDS Primary             │──Multi-AZ──► RDS Standby             │
│                          │  (auto)     │  (auto-failover)       │
│                          │             │                        │
│  ElastiCache Primary     │◄──repl──────►│  ElastiCache Replica  │
└──────────────────────────┘           └──────────────────────────┘
```

Cost Component	Monthly	Annual
Compute (same total, split across AZs)	(same base cost)	(same base cost)
RDS Multi-AZ standby	(same as Option A)	
Cross-AZ data transfer	**~$1,667**	**~$20,000**
Additional cost over single-AZ	**~$1,667/mo**	**~$20,000/yr**

AZ Failure RTO: < 2 minutes (automatic, traffic routes to surviving AZ) **Data Loss:** 0

5.3 Multi-AZ Comparison

Factor	Option A (Preferred Zone)	Option B (Active-Active)
Additional annual cost	~$3,000	~$20,000
AZ failure RTO	2-15 min	< 2 min
Cross-AZ latency	None (all traffic in 1 AZ)	1-2ms (imperceptible)
Complexity	Lower	Higher (pod topology constraints)
Recommendation	**Best value for this workload**	Justified only if < 2 min RTO is mandatory

> ***Andres's assessment is correct:*** *Option A with 2-3 minute downtime is the right call. AZ failures are rare (1-3 per region per year historically), and the $17K/yr savings outweighs the ~13 minute RTO difference.*

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6. Multi-Region DR Cost

6.1 Option A: Cold Standby (Daily Snapshots to us-east-1)

No running compute. Automated daily RDS snapshots copied cross-region.

Cost Component	Monthly	Annual
Cross-region snapshot transfer (~500 GB/day incremental × $0.02/GB)	~$50	~$600
Snapshot storage in us-east-1 (~2 TB total)	~$46	~$552
S3 cross-region replication (app data, configs)	~$20	~$240
Total DR cost	**~$116**	**~$1,392**

Regional Failure RTO: 3-6 hours (provision EKS, restore RDS from snapshot, deploy pods) **Data Loss (RPO):** Up to 24 hours (last snapshot)

6.2 Option B: Warm Standby (Standby RDS in us-east-1) — Rick's suggestion

Running RDS read replicas cross-region. No EKS compute.

Cost Component	Monthly	Annual
RDS cross-region read replicas (matches primary sizing)	~$2,500*	~$30,000
Cross-region replication data transfer	~$200	~$2,400
S3 cross-region replication	~$20	~$240
Total DR cost	**~$2,720**	**~$32,640**

*Estimate assumes 4-5 RDS instances (consolidated from 167 tenant DBs). Actual depends on instance sizing.

Regional Failure RTO: 1-2 hours (promote RDS replicas, provision EKS, deploy pods) **Data Loss (RPO):** 1-5 seconds (async replication lag)

6.3 Option C: Hybrid — Hot DB + Cold Compute (RECOMMENDED)

Maintain **one small RDS read replica** per database cluster in us-east-1 (not full-size — use smaller instance class). No EKS compute.

Cost Component	Monthly	Annual
RDS cross-region read replicas (smaller instance class)	~$800*	~$9,600
Cross-region replication data transfer	~$200	~$2,400
S3 cross-region replication	~$20	~$240
Total DR cost	**~$1,020**	**~$12,240**

*Uses db.r6g.large or db.t3.large instead of matching primary instance sizes.

Regional Failure RTO: 1-2 hours (promote replicas, scale up, provision EKS, deploy) **Data Loss (RPO):** 1-5 seconds **Advantage:** Checks the "multi-region" box for investors with minimal RPO, at 1/3 the cost of full-size replicas.

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7. Total Annual Cost — All Scenarios Combined

Compute + Multi-AZ (Preferred Zone) + Multi-Region DR

Compute Choice	Compute/yr	Multi-AZ	DR (Cold)	DR (Hybrid)	**Total w/ Cold DR**	**Total w/ Hybrid DR**
EC2 On-Demand	$199,704	$3,000	$1,392	$12,240	$204,096	$214,944
EC2 1yr RI	$123,588	$3,000	$1,392	$12,240	$127,980	$138,828
EC2 3yr RI	$80,316	$3,000	$1,392	$12,240	$84,708	$95,556
EC2 Right-sized 3yr RI	$66,876	$3,000	$1,392	$12,240	$71,268	$82,116
EC2 Hybrid (RI + Spot)	$103,752	$3,000	$1,392	$12,240	$108,144	$118,992
Fargate On-Demand	$143,244	$3,000	$1,392	$12,240	$147,636	$158,484
Fargate 1yr SP	$116,040	$3,000	$1,392	$12,240	$120,432	$131,280
Fargate 3yr SP	$82,044	$3,000	$1,392	$12,240	$86,436	$97,284

Including Estimated Operational Overhead

Scenario	Infra/yr	Ops Cost/yr	**True Total/yr**
EC2 1yr RI + Cold DR	$127,980	~$45,000	**~$173,000**
EC2 1yr RI + Hybrid DR	$138,828	~$45,000	**~$184,000**
EC2 3yr RI + Cold DR	$84,708	~$45,000	**~$130,000**
EC2 3yr RI + Hybrid DR	$95,556	~$45,000	**~$141,000**
EC2 Right-sized 3yr + Cold DR	$71,268	~$45,000	**~$116,000**
Fargate 1yr SP + Cold DR	$120,432	~$5,000	**~$125,000**
Fargate 1yr SP + Hybrid DR	$131,280	~$5,000	**~$136,000**
Fargate 3yr SP + Cold DR	$86,436	~$5,000	**~$91,000**
Fargate 3yr SP + Hybrid DR	$97,284	~$5,000	**~$102,000**

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8. AZ Failure Frequency — Historical Context

Year	Notable AWS AZ/Region Incidents	Region
2023	Multiple us-east-1 incidents; us-west-2 incident (Sep)	us-east-1, us-west-2
2024	us-east-1 Kinesis failure (Jul, ~7 hrs)	us-east-1
2025	eu-north-1 Stockholm (Feb); us-east-1 (Oct, ~15 hrs); us-west-2 (Oct)	Multiple

Key takeaway:

- **Full AZ failures:** ~1-3 per region per year, lasting minutes to hours
- **Full region failures:** Extremely rare (~once every few years, typically us-east-1)
- **us-west-2 (Oregon):** Historically more stable than us-east-1, fewer major incidents
- **Conclusion:** The "Preferred Zone + Warm Standby" architecture with 2-5 min AZ failover RTO is well-justified. The ~$17K/yr savings over active-active is worth it given AZ failures happen ~1-3 times/year.

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9. Recommendation

Best Architecture: Preferred Zone + Warm Standby (Multi-AZ) with Hybrid DR

```
us-west-2a (PRIMARY)                    us-west-2b (STANDBY)
┌──────────────────────────┐            ┌──────────────────────┐
│  Full EKS Cluster        │            │  2-3 Warm Spare      │
│  EC2 or Fargate          │            │  Nodes (autoscale)   │
│                          │            │                      │
│  RDS Primary             │──Multi-AZ──▶ RDS Standby          │
│  ElastiCache Primary     │            │  ElastiCache Replica  │
└──────────────────────────┘            └──────────────────────┘

us-east-1 (COLD DR)
┌──────────────────────────────────────┐
│  RDS Read Replicas (small instances) │
│  No EKS (provision on demand)        │
│  S3 cross-region replication         │
│  Daily snapshot backups              │
└──────────────────────────────────────┘
```

Best Compute Choice: Depends on Team Capacity

If you have (or will hire) dedicated K8s/DevOps capacity:

EC2 with 1yr Reserved Instances + Spot for delayed-job workers

Component	Annual
EC2 compute (RI + Spot)	$103,752
Multi-AZ (preferred zone)	$3,000
Hybrid DR (us-east-1)	$12,240
Ops overhead (~0.3 FTE)	$45,000

Total	**~$164,000/yr**

Pros: Most cost-effective on paper, full control, Spot savings on workers **Cons:** Requires ongoing node management expertise, OS patching, AMI updates

If you prefer minimal operational burden (RECOMMENDED for lift-and-shift):

Fargate with 1yr Compute Savings Plan

Component	Annual
Fargate compute (1yr SP)	$116,040
Non-Fargate infra (ElastiCache, monitoring EC2)	$6,360
Multi-AZ (preferred zone)	$3,000
Hybrid DR (us-east-1)	$12,240
Ops overhead (minimal)	$5,000
Total	**~$143,000/yr**

Pros: Zero node management, auto-scaling per pod, simpler operations, faster time-to-migrate **Cons:** Need to refactor DaemonSets to sidecars, VPC subnet sizing for 1,141 ENIs, no EBS

Best value long-term (3yr commitment):

Fargate with 3yr Compute Savings Plan

Component	Annual
Fargate compute (3yr SP)	$82,044
Non-Fargate infra	$6,360
Multi-AZ	$3,000
Hybrid DR	$12,240
Ops overhead	$5,000
Total	**~$109,000/yr**

Decision Matrix

Factor	EC2	Fargate	Winner
Raw compute cost (1yr)	$123,588	$116,040	**Fargate** (surprisingly)
Raw compute cost (3yr)	$80,316	$82,044	EC2 (barely)
Operational overhead	~$45K/yr	~$5K/yr	**Fargate**
Total cost of ownership	~$164K/yr	~$143K/yr	**Fargate**
Time to migrate	Longer (node config)	Faster (Fargate profiles)	**Fargate**

DaemonSet support	Yes	No	EC2
EBS/StatefulSet support	Yes	No (EFS only)	EC2
Debugging/SSH access	Full	Limited	EC2
Spot instance savings	Yes (workers)	No	EC2
Auto-scaling granularity	Node-level	Pod-level	**Fargate**
Overall for lift-and-shift	Good	**Better**	**Fargate**

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10. Practical Migration Path

If choosing Fargate:

1. **Mixed compute strategy:** Fargate for all stateless pods (~95% of workload), small EC2 managed node group (2-3 nodes) for monitoring DaemonSets
2. **Replace in-cluster stateful services:** Memcached → ElastiCache, Redis → ElastiCache, Prometheus → Amazon Managed Prometheus or EC2 node
3. **Fargate profiles:** One per namespace pattern (tenant namespaces, services, prex, monitoring)
4. **VPC planning:** /16 CIDR minimum to accommodate 1,141+ ENIs across 2 AZs
5. **Resource requests mandatory:** Every pod MUST have CPU/memory requests defined (Fargate requires this)

If choosing EC2:

1. Use Karpenter (not Cluster Autoscaler) for intelligent node provisioning
2. Standard pool: 1yr RI (committed baseline), Karpenter on-demand for burst
3. Delayed-job pool: Spot instances with on-demand fallback
4. Consider Graviton (m7g) instances for ~20% better price/performance (requires ARM image builds)

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11. RDS Cost Estimate (Both Architectures Need This)

Component	Instance	Multi-AZ Monthly	Annual
Primary MariaDB (tenant DBs)	db.r6g.2xlarge	~$1,800	~$21,600
Prex PostgreSQL (prod)	db.r6g.large	~$400	~$4,800
Prex PostgreSQL (staging)	db.t3.medium	~$140	~$1,680
RDS subtotal		**~$2,340**	**~$28,080**
DR replicas in us-east-1 (small)	db.r6g.large × 2 + db.t3.medium	~$800	~$9,600

> *Note: RDS cost is the same regardless of EC2 vs Fargate compute choice.*

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12. Grand Total Estimate

Recommended: Fargate 1yr SP + Preferred Zone + Hybrid DR

Line Item	Monthly	Annual

Fargate compute (1yr Savings Plan)	$9,670	$116,040
ElastiCache (Memcached + Redis)	$180	$2,160
EC2 monitoring nodes (3x t3.medium)	$91	$1,092
RDS Primary (Multi-AZ)	$2,340	$28,080
RDS DR replicas (us-east-1)	$800	$9,600
EKS control plane	$73	$876
Multi-AZ warm spare nodes	$173	$2,076
Cross-region data transfer	$250	$3,000
NAT Gateway (2 AZs)	$130	$1,560
Application Load Balancer	$50	$600
ECR storage + transfer	$50	$600
Route 53 + health checks	$30	$360
Infrastructure Total	**$13,837**	**$166,044**
Operational overhead (minimal)	$417	$5,000
Grand Total	**~$14,254**	**~$171,044**

Alternative: EC2 1yr RI + Spot Workers + Preferred Zone + Hybrid DR

Line Item	Monthly	Annual
EC2 compute (RI + Spot)	$8,646	$103,752
ElastiCache (Memcached + Redis)	$180	$2,160
RDS Primary (Multi-AZ)	$2,340	$28,080
RDS DR replicas (us-east-1)	$800	$9,600
EKS control plane	$73	$876
Multi-AZ warm spare nodes	$173	$2,076
Cross-region data transfer	$250	$3,000
NAT Gateway (2 AZs)	$130	$1,560
Application Load Balancer	$50	$600
ECR storage + transfer	$50	$600
Route 53 + health checks	$30	$360
Infrastructure Total	**$12,722**	**$152,664**

Operational overhead (~0.3 FTE)	$3,750	$45,000
Grand Total	**~$16,472**	**~$197,664**

Pricing based on publicly available AWS rates as of March 2026. Actual costs may vary based on usage patterns, data transfer volumes, and negotiated discounts. All prices are for us-west-2 (Oregon). Recommend validating with AWS Pricing Calculator before finalizing.

Sources: AWS Fargate Pricing, EC2 On-Demand Pricing, EC2 Instance Comparison (Vantage), RDS Pricing, EKS Pricing, Compute Savings Plans, Rafay EC2 vs Fargate Cost Comparison